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Exercise Workbook

Introduction to Electric Power Conversion

version 2025.00

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Introduction

Electric power conversion is a central theme in the technological development of electrical engineering disciplines and represents one of the cornerstones of the ongoing energy transition. The growing integration of renewable sources, the electrification of transport and the advancement of storage systems make this subject not only highly relevant but also strategic for the education of future engineers. Given this importance, a wide range of textbooks and resources are available that thoroughly address the theoretical and practical aspects of electric energy conversion.

In recent years, teaching methods have been further enriched by the availability of video lectures and blended learning approaches, which provide flexibility and immediacy in accessing content. Digital platforms undoubtedly offer significant advantages, such as the possibility of monitoring which parts of a resource have been effectively viewed or revisited by students, thus providing useful feedback to instructors. However, these tools also involve certain risks: the learning process may be perceived as reduced to simply watching a multimedia resource — at times in accelerated mode — without truly engaging the student. Numerous pedagogical approaches emphasize that learning is more effective when it actively engages multiple senses and requires direct effort from the learner. In this respect, the passive consumption of a multimedia resource may risk “passivizing” the student, limiting the depth of the educational experience.

This workbook has been conceived precisely with the aim of fostering active student engagement through a series of exercises designed to guide learners in the analysis and calculation of concrete cases. Drawing graphs, performing calculations by hand and developing diagrams with pen and paper complement computer-based simulations, creating a truly in-

tegrated approach. The underlying conviction is that the combination of traditional and digital tools provides undeniable educational benefits: on the one hand, it enhances the understanding of the concepts covered, and on the other, it supports long-term retention and assimilation of the most important concepts.

This workbook assumes prior knowledge of basic electrical engineering, in particular the ability to analyze electrical circuits both in steady-state conditions and in periodic (general) regimes. Short theoretical notes are provided throughout the text, but readers are referred to specialized textbooks for a more comprehensive treatment of the underlying theory.

The focus is limited to static conversion circuits, commonly referred to as “power electronics”. The topics covered are intended as a first introduction to the subject and therefore only fundamental configurations are presented and discussed. For numerical simulations, the software PLECS is suggested, and a short introduction to its use is provided in the appendix.

Further improvements to the text are under development and I will be therefore grateful to anyone who would like to report errors, suggestions or inaccuracies to me at luigi.alberti@unipd.it.

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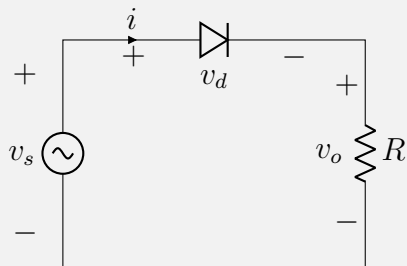
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CHAPTER 2

Half-wave rectifiers

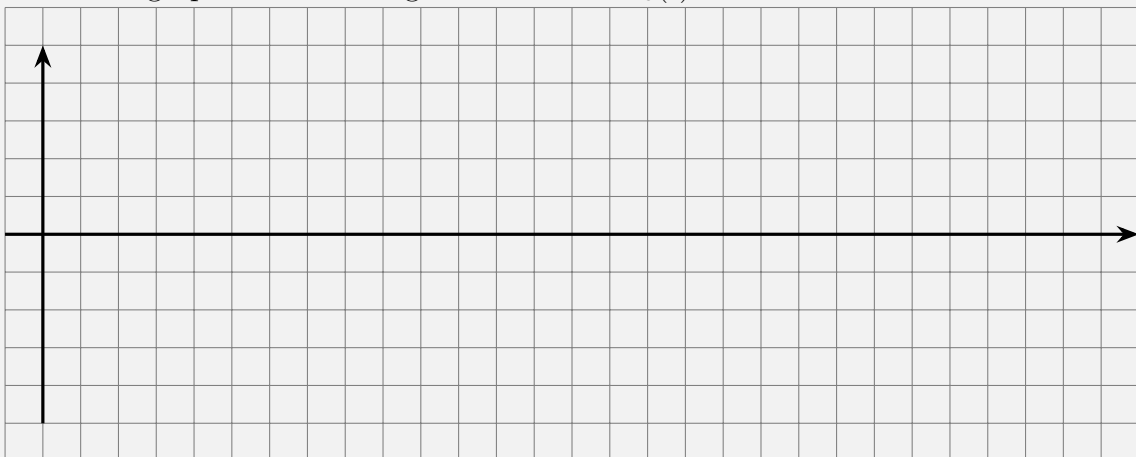
Exercise 2.1: half-wave rectifier with resistive load.



The half-wave rectifier in the figure supplies a resistor $R=15\ \Omega$. The generator parameters are:

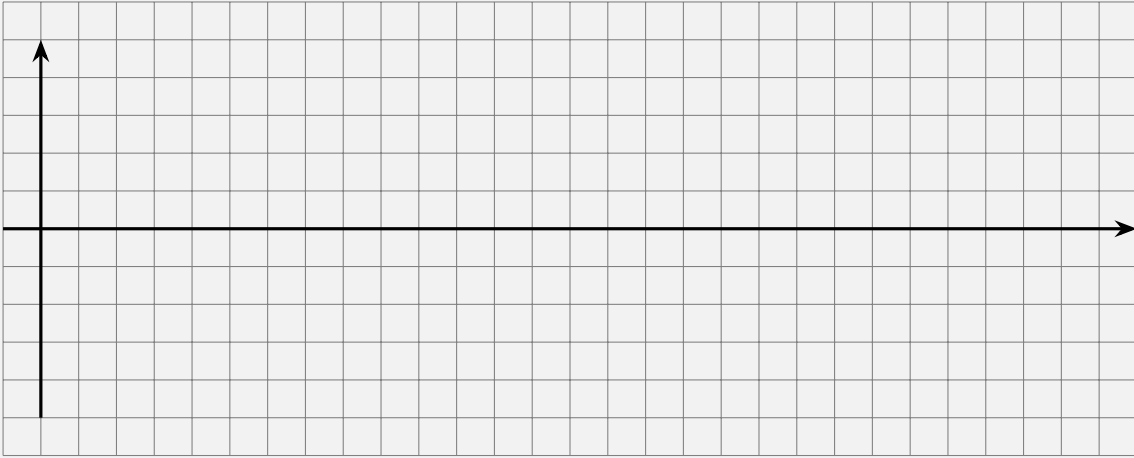
$$v_s = \sqrt{2} \cdot 200 \sin(2\pi ft)$$
$$f = 50\ \text{Hz}$$

a) Draw the graph for the voltage of the source $v_s(t)$.

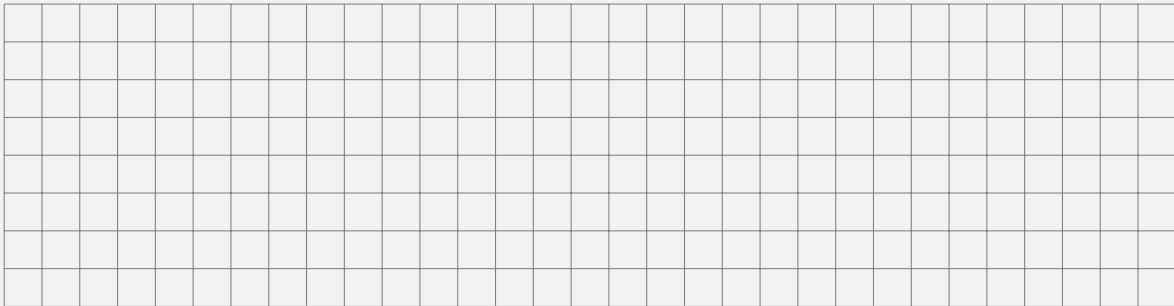


2.1

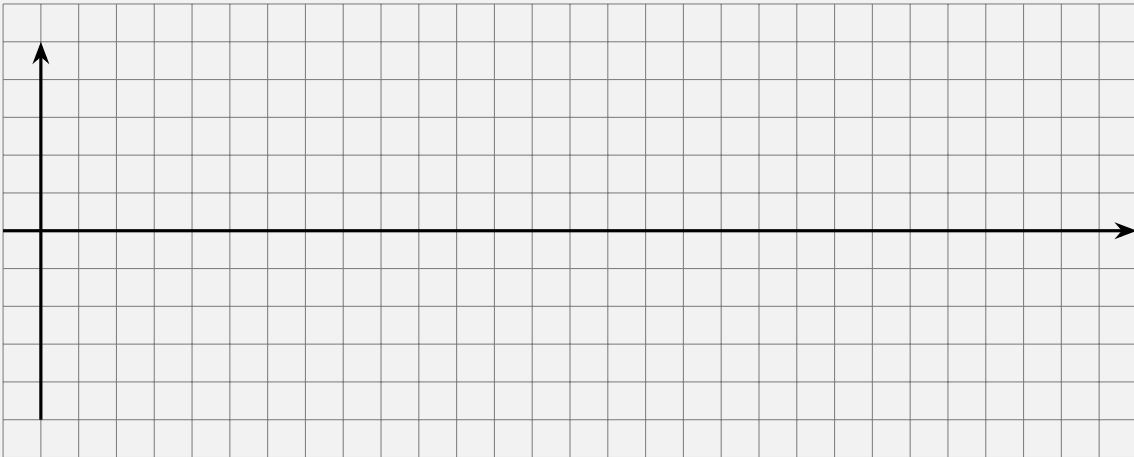
b) Draw the graph for the output voltage $v_o(t)$.



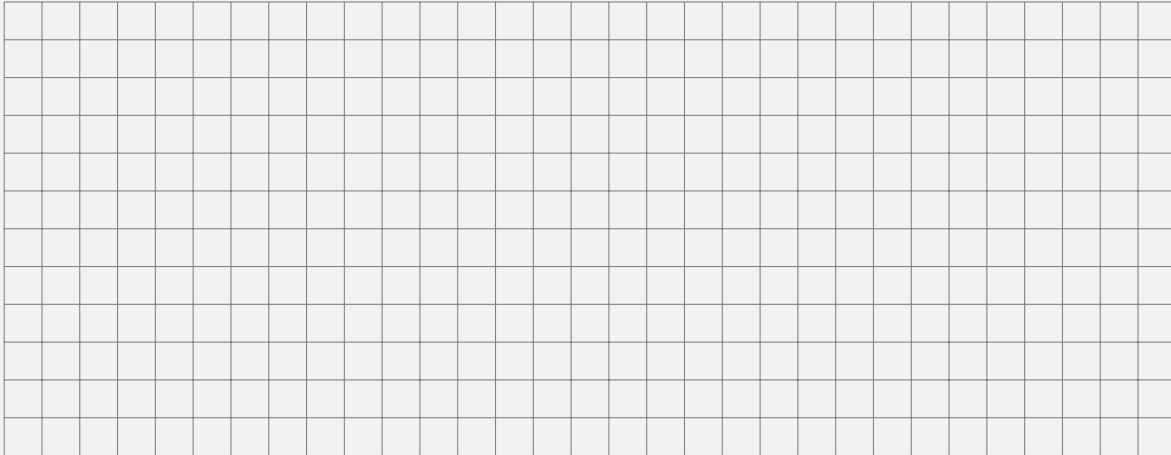
c) Compute the average and rms value of the output voltage $v_o(t)$.



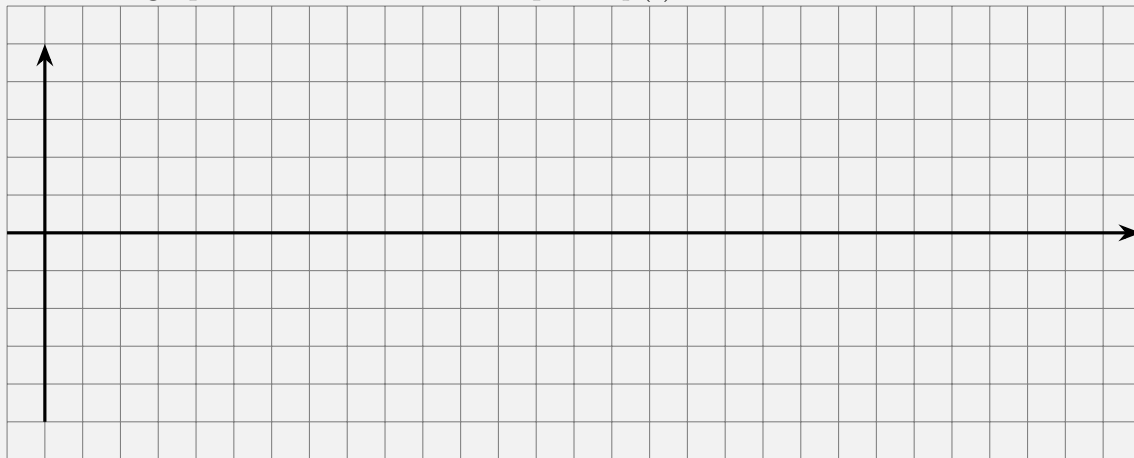
d) Draw the graph for the circuit current $i(t)$.



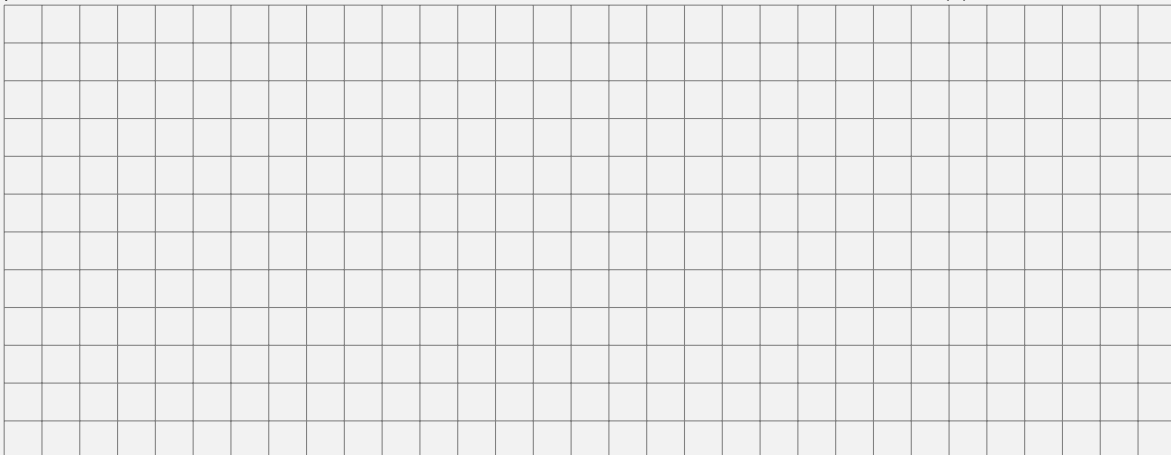
e) Compute the average and rms value of the circuit current $i(t)$.



f) Draw the graph for the instantaneous power $p(t)$ on the resistor.



g) Compute the average power and the power factor for the source $v_s(t)$.

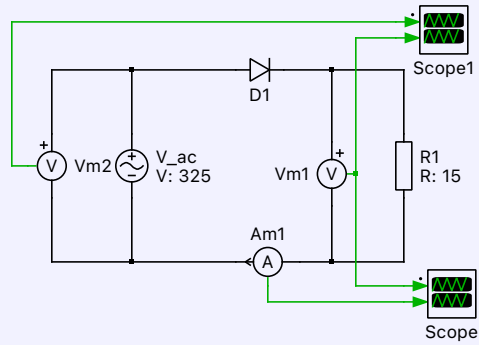




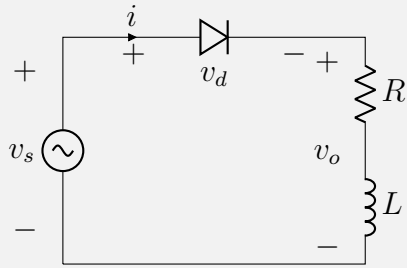
2.1

Lab. 2.1: half-wave rectifier with resistive load.

Build a PLECS model to simulate the Exercise 2.1 and compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



Exercise 2.2: half-wave rectifier with resistive and inductive load.

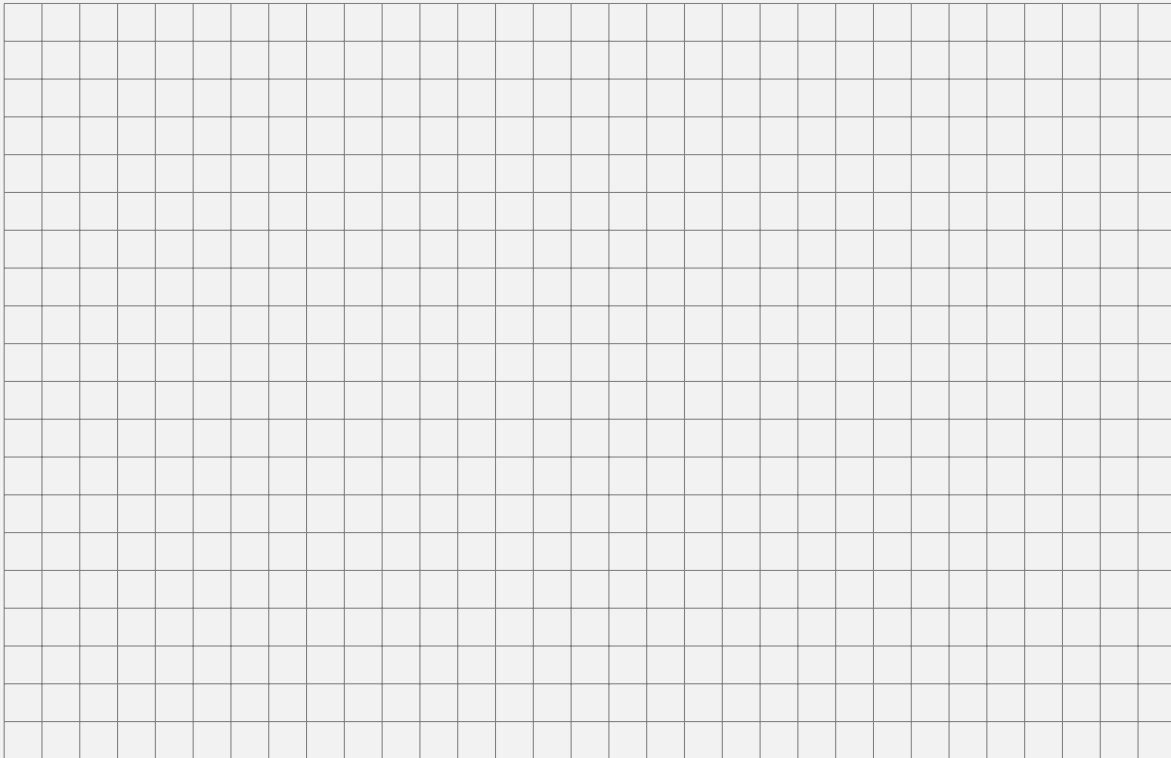


The half-wave rectifier in the figure supplies a resistor $R=30\ \Omega$ and an inductor $L=0.150\ \text{H}$. The generator parameters are:

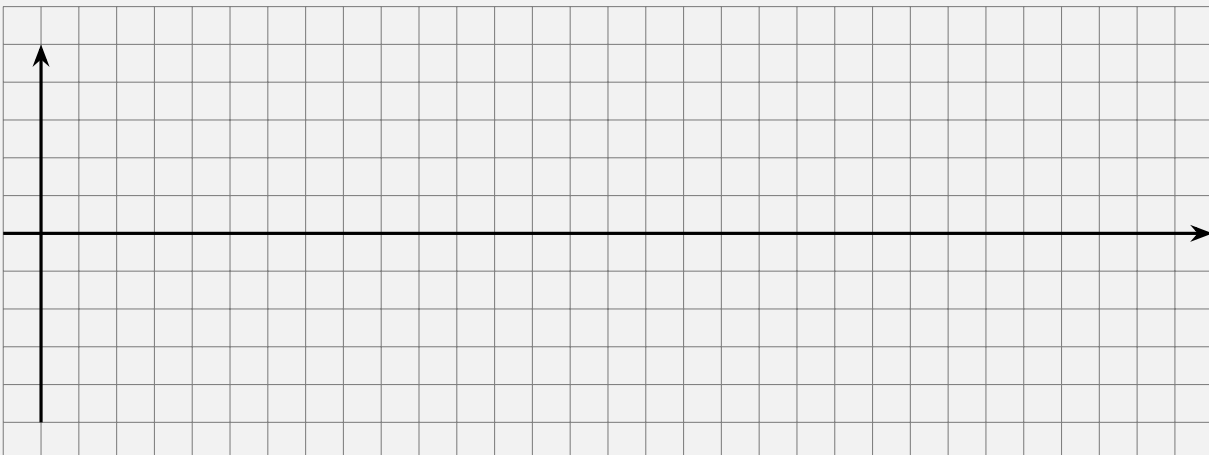
$$v_s = 100 \sin(2\pi ft)$$

$$f = 50\ \text{Hz}$$

- a) Compute the extinction angle β and write the expression for the current $i(t)$.

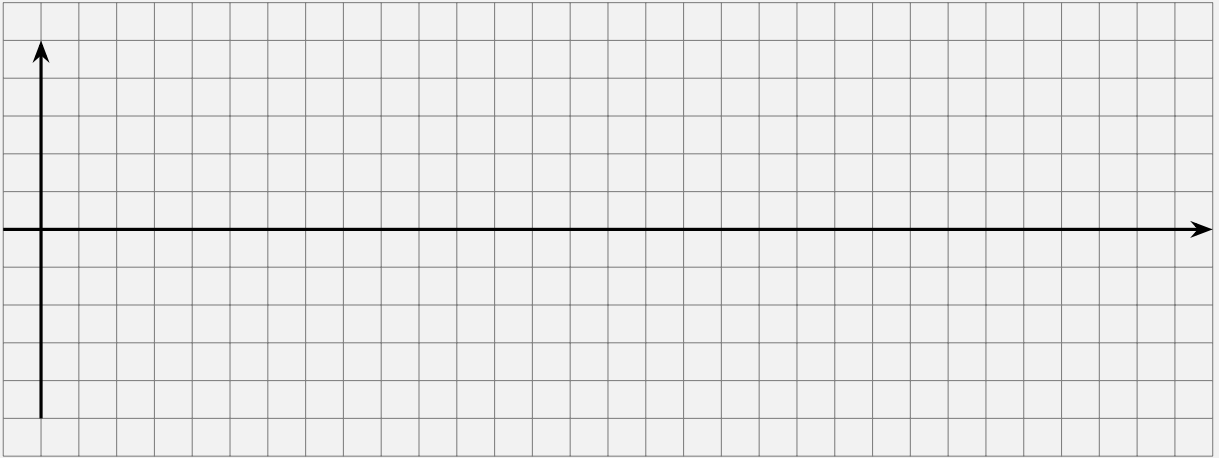


- b) Draw the graph for the voltage for the source $v_s(t)$ and sketch the current $i(t)$.

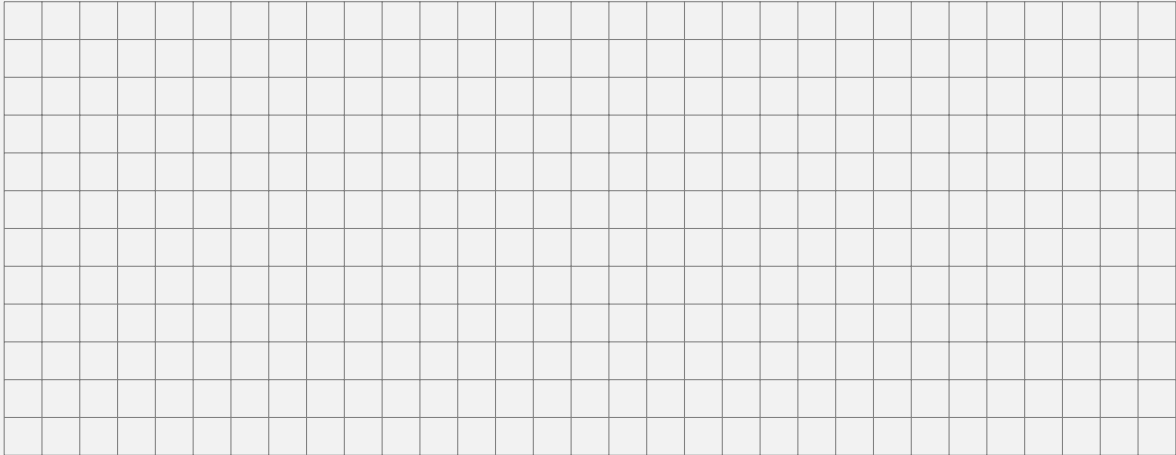


2.2

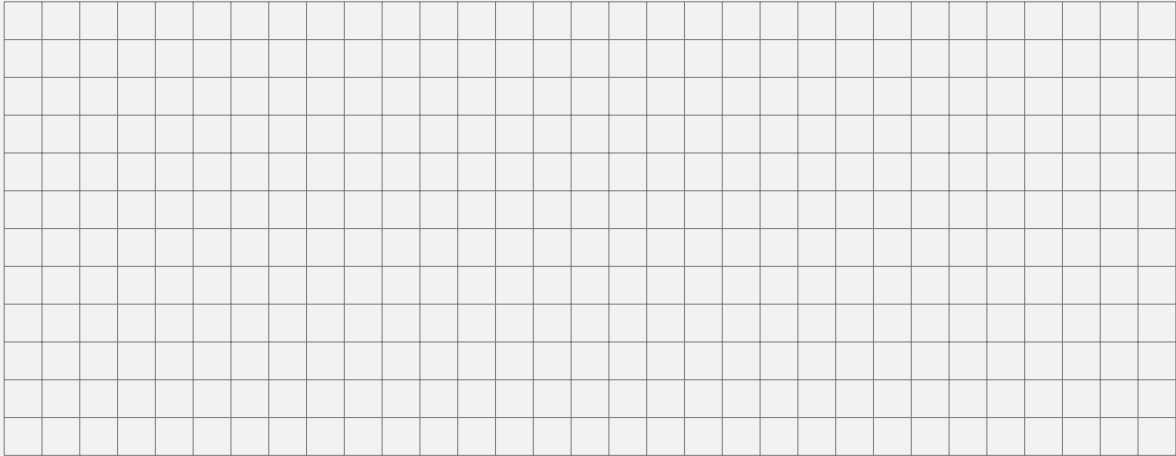
c) Draw the graph for the output voltage $v_o(t)$.



d) Compute the average value of the output voltage $v_o(t)$.



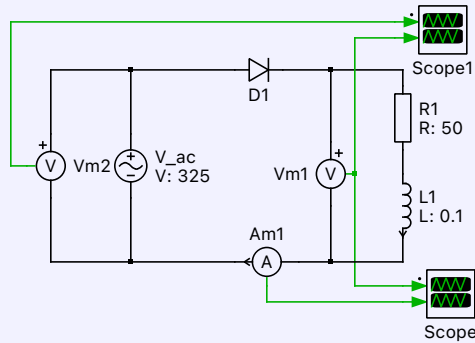
e) Compute the average value of the circuit current $i(t)$.



Lab. 2.2: half-wave rectifier with resistive and inductive load.

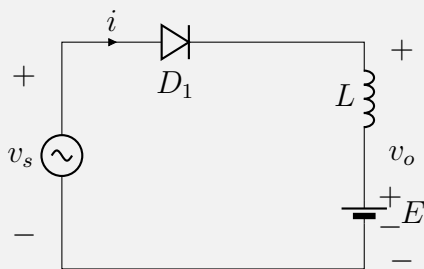
2.2

Build a PLECS model to simulate the Exercise 2.2 and compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



Exercise 2.3: half-wave rectifier with inductive and battery load.

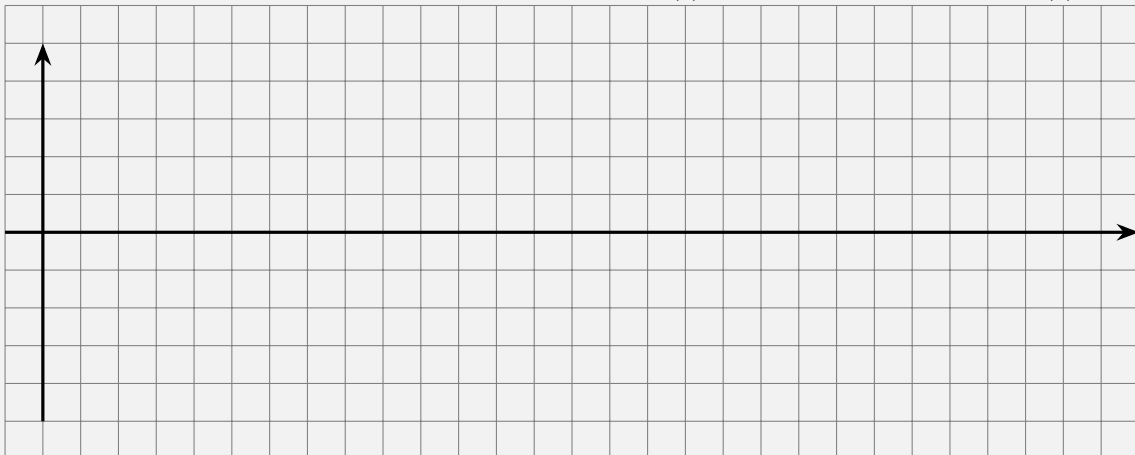
2.3



The half-wave rectifier in the figure supplies an inductor $L=0.01$ H in series with a battery $E=100$ V. The generator parameters are:

$$v_s = V_m \sin(2\pi ft) = 325 \cdot \sin(2\pi 50 \cdot t)$$

- a) Draw the graph for the voltage for the source $v_s(t)$ and sketch the current $i(t)$.



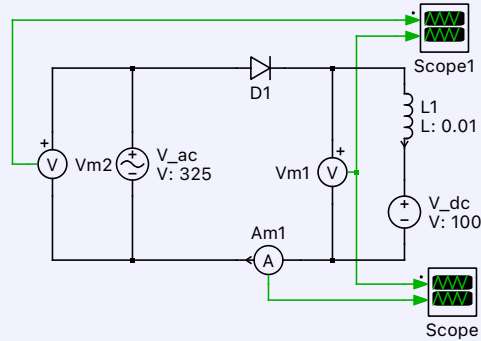
b) write an expression for the current in the circuit $i(t)$.

[illegible][illegible][illegible]

Lab. 2.3: half-wave rectifier with inductive and battery load.

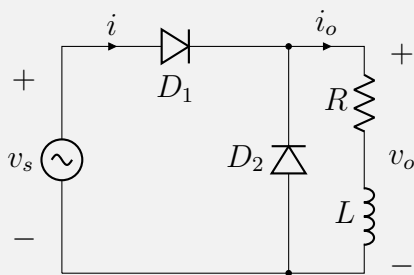
2.3

Build a PLECS model to simulate the Exercise 2.3 and compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



Exercise 2.4: half-wave rectifier with free-wheeling diode.

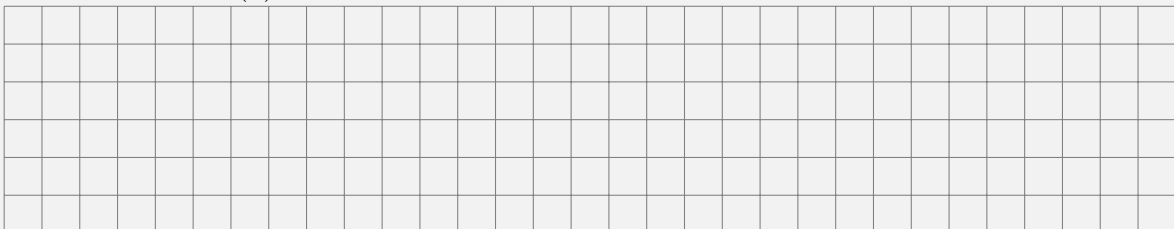
2.4



The half-wave rectifier in the figure supplies a resistor $R=5\ \Omega$ and an inductor $L=0.030\ \text{H}$. The generator parameters are:

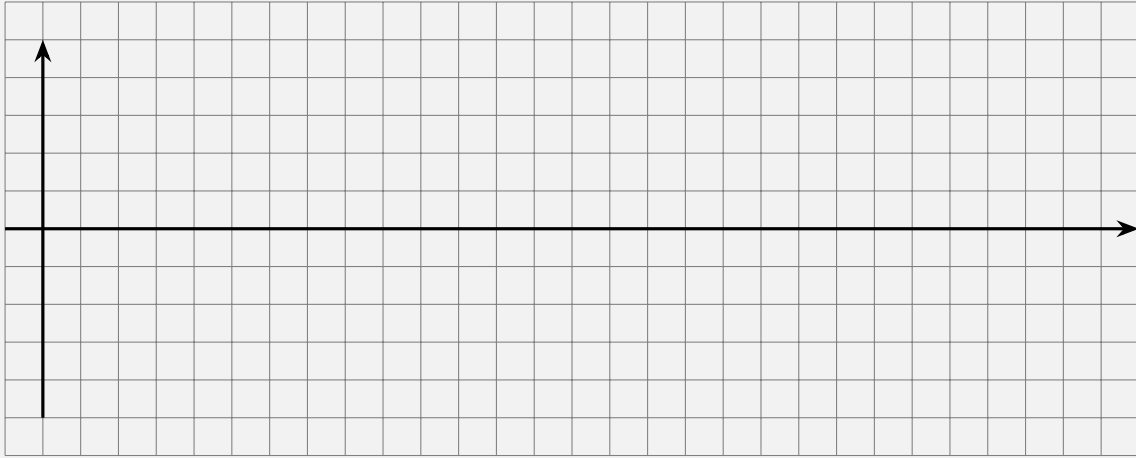
$$v_s = V_m \sin(2\pi ft) = 200 \cdot \sin(2\pi \cdot 50 \cdot t)$$

- a) Estimate the time necessary for the circuit to reach steady-state starting from a zero initial condition $i_o(0) = 0$.

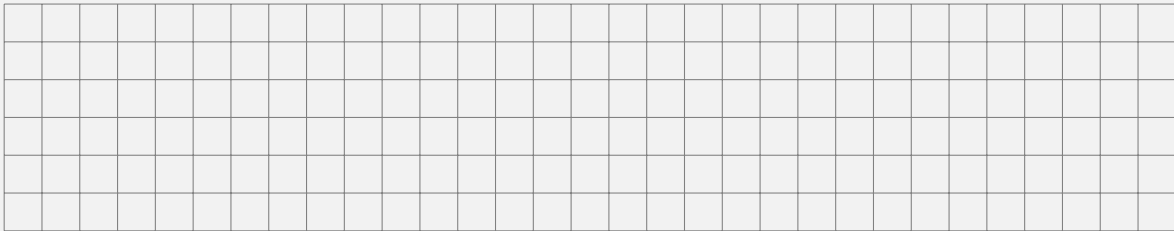


2.4

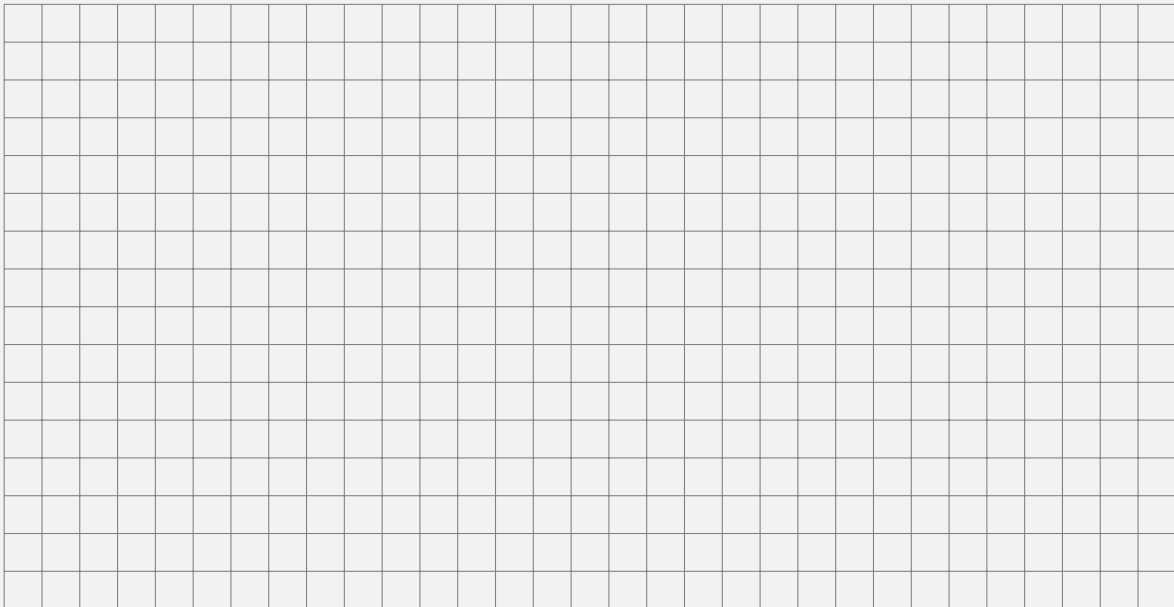
b) Draw the graph for the output voltage $v_o(t)$.



c) Compute the average voltage and current in the load.

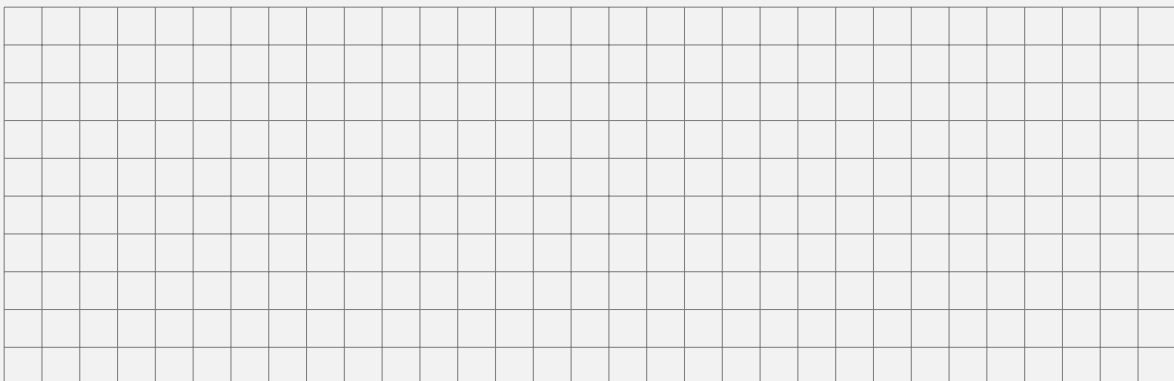
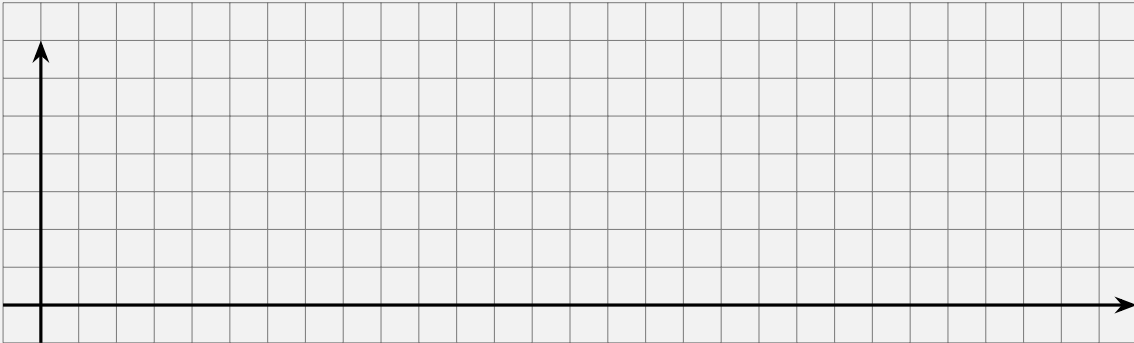
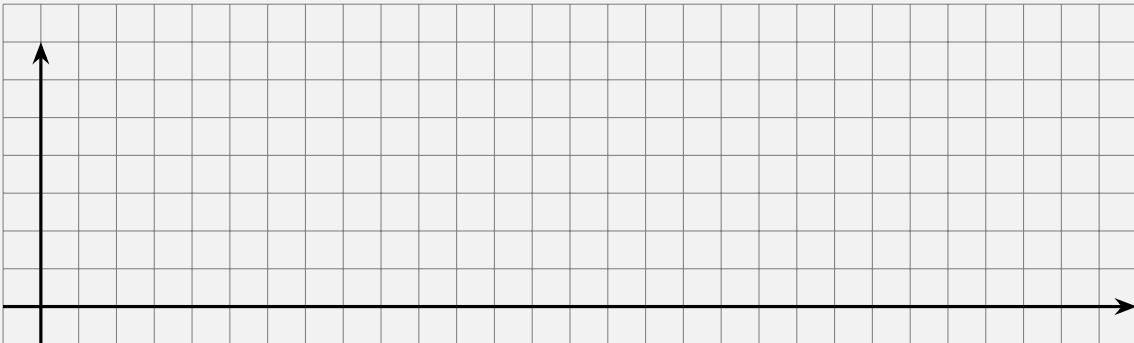
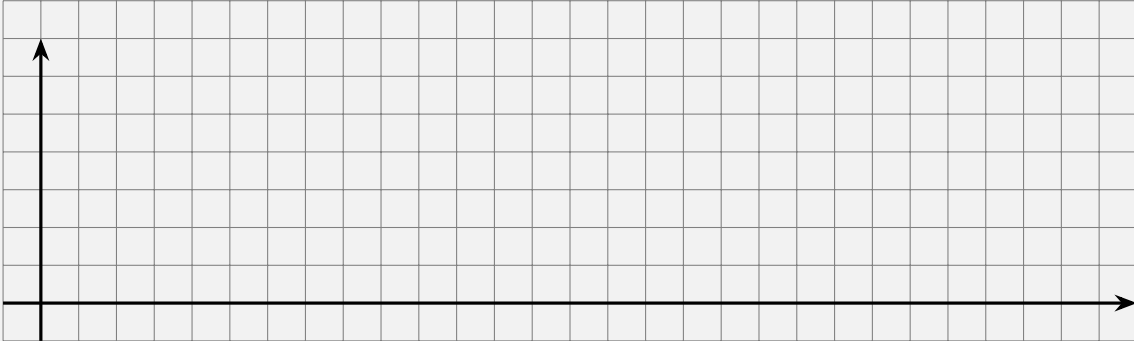


d) Compute the power delivered to the load.



2.5

d) Assuming an infinite inductance, draw the graph for the generator current i_s , in the two diodes i_{D1} and i_{D2} , and compute the power factor of v_s .

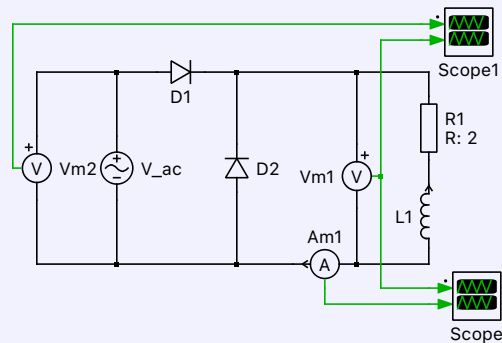


Lab. 2.4: half-wave rectifier with free-wheeling diode.



2.4

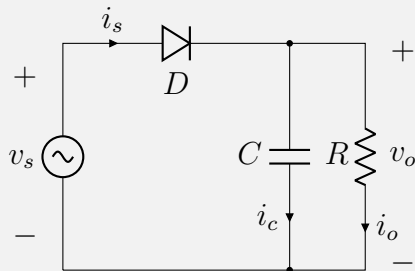
Build a PLECS model to simulate the Exercise 2.4 and 2.5. Compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



Exercise 2.6: half-wave rectifier with capacitor filter.



2.6



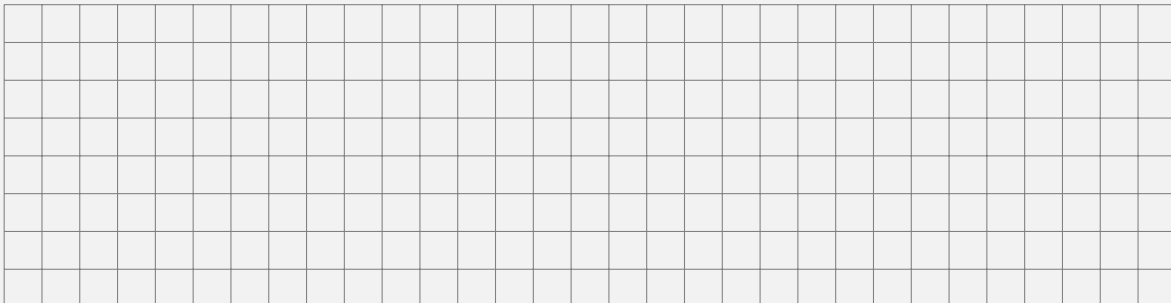
$$v_s = V_m \cdot \sin(2\pi ft)$$

$$V_m = 325$$

$$f = 50Hz$$

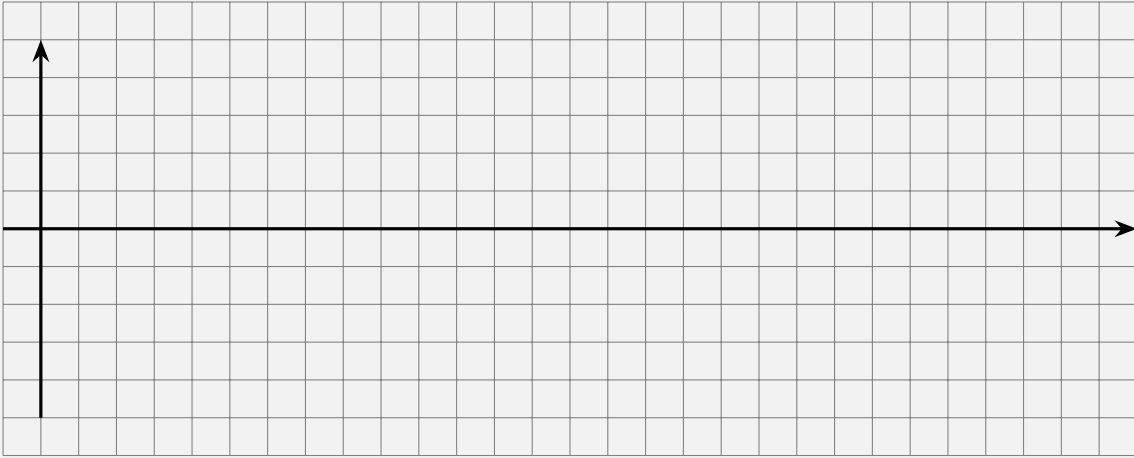
$$R = 100\Omega$$

a) Compute the capacitor value to achieve an output voltage ripple $\Delta v_o \leq 0.1V_m$

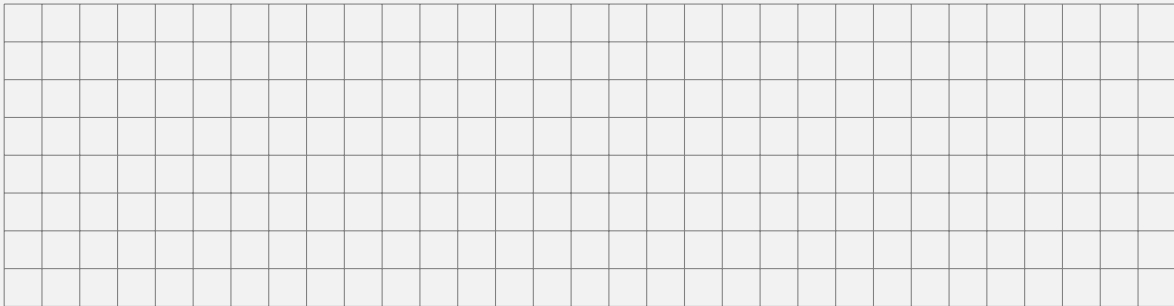


2.6

b) Sketch an approximated graph for the output voltage $v_o(t)$ highlighting Δv_o .



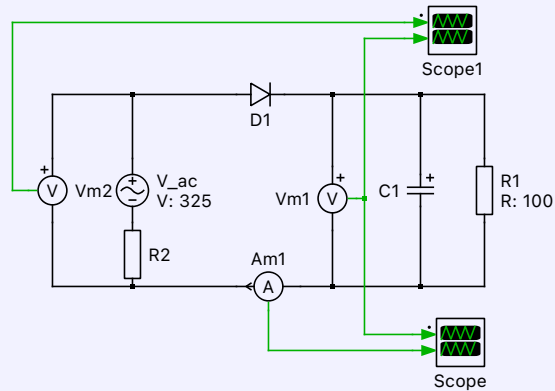
c) Compute an estimation for the average output voltage V_o



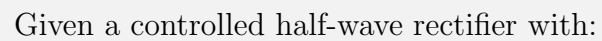
2.5

Lab. 2.5: half-wave rectifier with capacitor filter.

Build a PLECS model to simulate the Exercise 2.6 and compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:



Q 2.7



$$f = 50Hz$$

a) Compute the phase angle α to adopt for controlling the rectifier to obtain and average output voltage $V_o=50$ V.

[illegible]

b) Compute the power delivered to the load.

[illegible]

c) Compute the power factor for the generator v_s .

[illegible]



Exercise 2.8: controlled half-wave rectifier with specified power.

Considering again the rectifier of Exercise 2.7

- a) Compute the angle α to achieve an output power $P = 200$ W.

This image shows a full page of blank graph paper. The background is a very light gray, and it is covered by a grid of thin, darker gray lines. The grid consists of small, equal-sized squares that extend across the entire visible area of the page. There are no margins, text, or other markings on the paper.

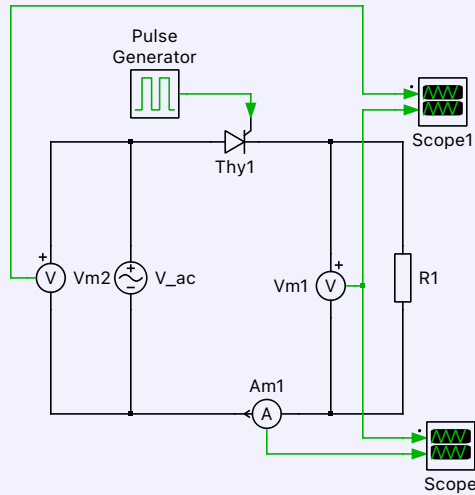
- b)** Is it possible to obtain a power $P = 250$ W in the resistor? Motivate the answer.

This image shows a full page of blank graph paper. The background is a very light gray, and it is covered by a grid of thin, darker gray lines. The grid consists of small, equal-sized squares that extend across the entire width and height of the page, leaving a small margin at the top. There are no markings, text, or drawings on the grid itself.

Lab. 2.6: controlled half-wave rectifier.

2.6

Build a PLECS model to simulate the Exercise 2.7 and 2.8. Compare the simulation results with the analytical calculations. Consider the following PLECS model to start the simulation:

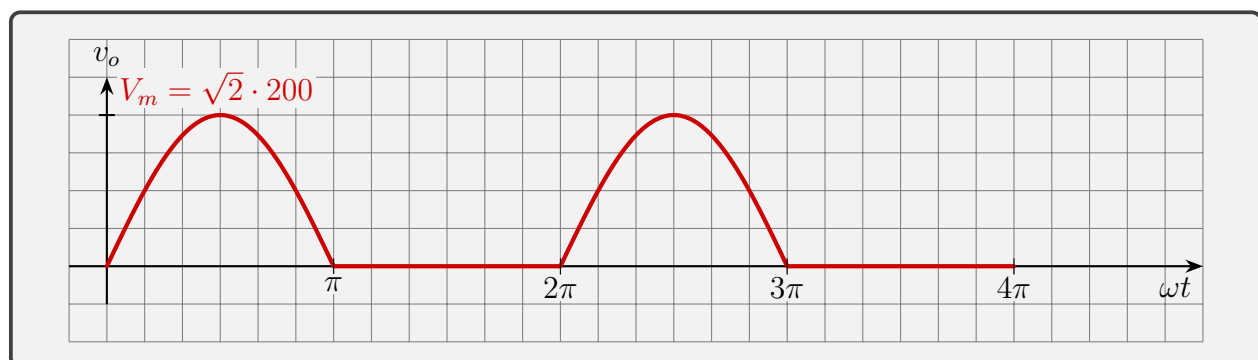
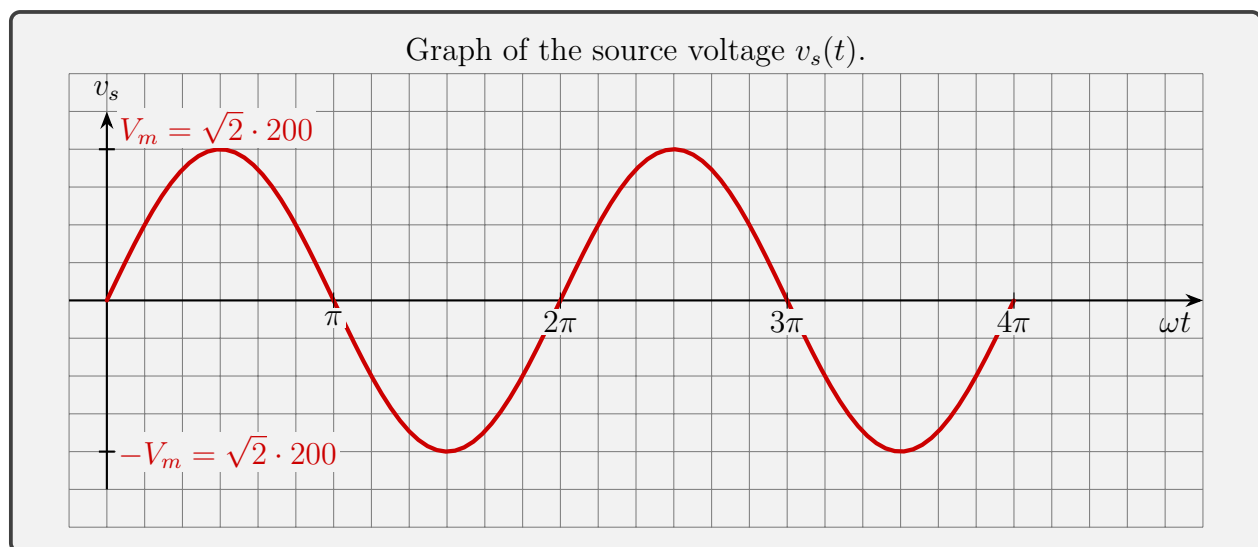


Before performing the computations in PLECS check that the output voltage waveform is as expected, in particular check if a proper phase delays (with the correct unit of measurements) has been set for the Pulse Generator block driving the SCR.

CHAPTER 6

Half-wave rectifiers

6.1 Result of exercise 2.1 on page 31



2.1 c

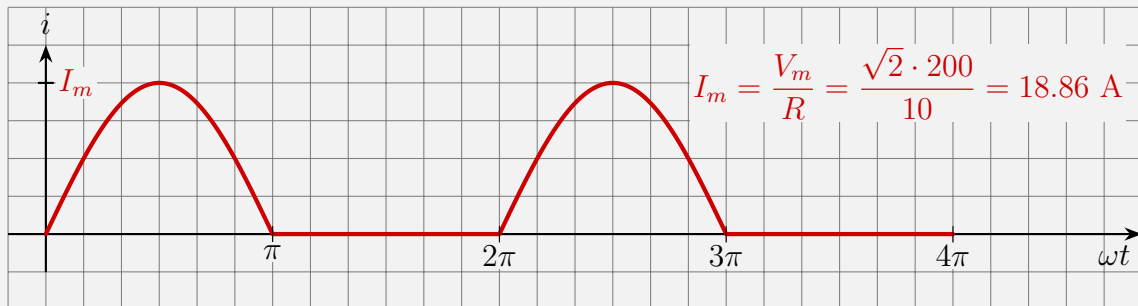
The average value of the output voltage $v_o(t)$ is:

$$V_o = \frac{V_m}{\pi} = \frac{\sqrt{2} \cdot 200}{\pi} = \frac{300}{\pi} = 24 \text{ V}$$

The rms value of the output voltage $v_o(t)$ is:

$$V_{o,rms} = \frac{V_m}{2} = \frac{\sqrt{2} \cdot 200}{2} = 141.42 \text{ V}$$

2.1 d



2.1 e

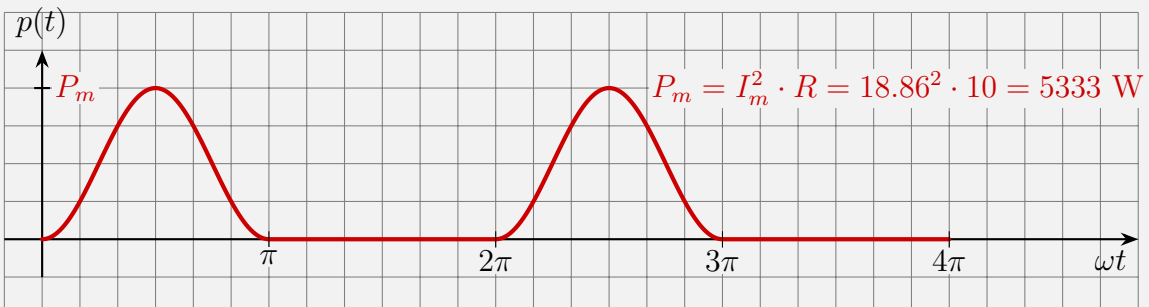
The average value of the output current $i(t)$ is:

$$I_o = \frac{V_o}{R} = \frac{24}{10} = 19.24 \text{ A}$$

The rms value of the current $i(t)$ is:

$$I_{rms} = \frac{V_{o,rms}}{R} = \frac{141.42}{10} = 22.51 \text{ A}$$

2.1 f



The average power in the source is the same as in the load and is computed as:

$$P = \frac{V_{rms}^2}{R} = R \cdot I_{rms}^2 = 1333 \text{ W}$$

The power factor is computed as:

$$pf = \frac{P}{S} = \frac{P}{\frac{V_m}{\sqrt{2}} I_{rms}} = \frac{1333}{200 \cdot 22.51} = 0.707$$

2.1 g)

Personal notes:

6.2 Result of exercise 2.2 on page 35

2.2 a

With:

$$Z = \sqrt{R^2 + (wL)^2} = \sqrt{30^2 + (2\pi 50 \cdot 0.150)^2} = 56 \, \Omega$$

$$\varphi = \arctan \frac{2\pi f L}{R} = \arctan \frac{2\pi 50 \cdot 0.150}{30} = 1.004 \, \text{rad} = 57.52^\circ$$

$$w\tau = 2\pi f \frac{L}{R} = 2\pi 50 \frac{0.150}{30} = 1.571 \, \text{rad}$$

The current can be written as:

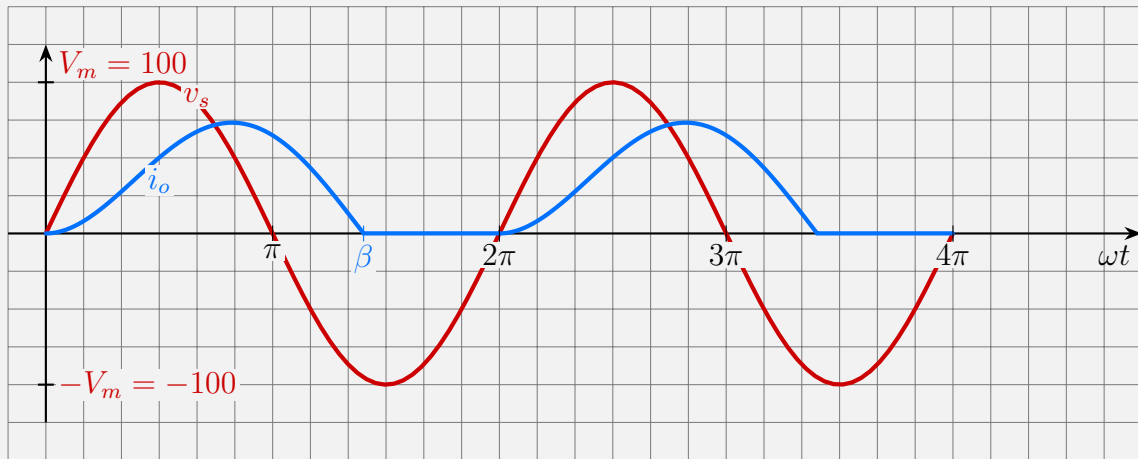
$$\begin{aligned} i(\omega t) &= \frac{V_m}{Z} [\sin(\omega t - \varphi) + \sin(\varphi) \cdot e^{-\omega t/\omega\tau}] \\ &= 1.79 [\sin(\omega t - 1.004) + \sin(1.004) \cdot e^{-\omega t/1.571}] \\ &\text{for } 0 \leq \omega t \leq \beta \end{aligned}$$

The extinction angle β is computed solving the following equation:

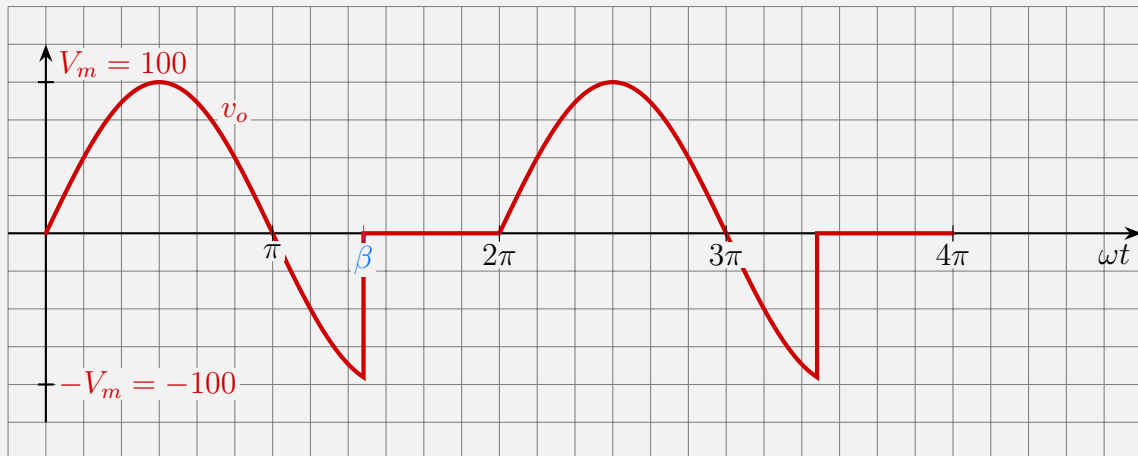
$$0 = [\sin(\beta - 1.004) + \sin(1.004) \cdot e^{-\beta/1.571}] = f(\beta)$$

Starting from π , with some trials an approximation is computed as $\beta = 4.204 \, \text{rad}$.

2.2 b



Personal notes:



2.2 c)

The average voltage at the output is computed as:

$$\begin{aligned} V_o &= \frac{1}{2\pi} \int_0^\beta v_o(\omega t) d(\omega t) \\ &= \frac{V_m}{2\pi} [1 - \cos(\beta)] \\ &= 24 \text{ V} \end{aligned}$$

2.2 d)

The average current in the load is:

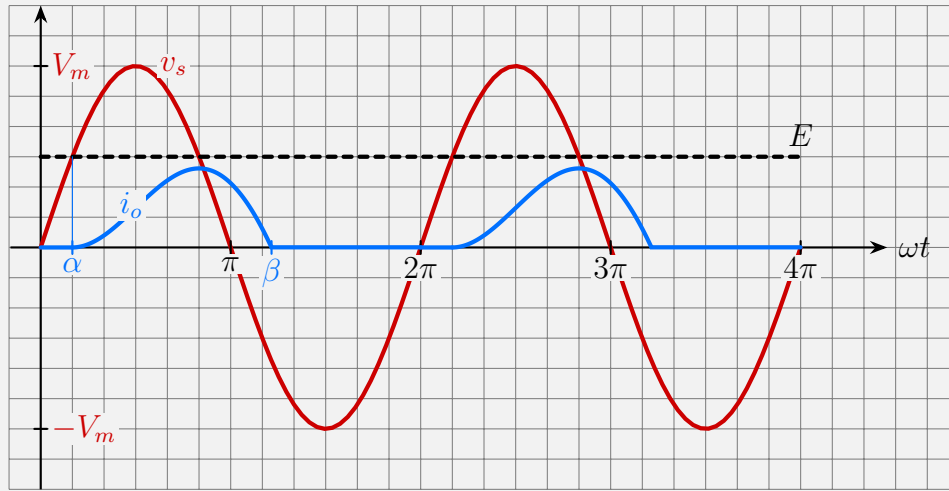
$$I_o = \frac{V_o}{R} = \frac{24}{30} = 0.79 \text{ A}$$

2.2 e)

Personal notes:

6.3 Result of exercise 2.3 on page 37

2.3 a



2.3 b

The diode starts to conduct at $\omega t = \alpha$:

$$\alpha = \arcsin \frac{E}{V_m} = \arcsin \frac{150}{300} = 60 \text{ rad}$$

for $\alpha \leq \omega t \leq \beta$ the current can be expressed as:

$$\begin{aligned} i(\omega t) &= \frac{V_m}{\omega L} \left\{ \cos(\alpha) - \cos(\omega t) \right\} + \frac{E}{\omega L} (\alpha - \omega t) \\ &= \frac{300}{314.159 \cdot 0.050} \left\{ \cos(60) - \cos(314.159 \cdot t) \right\} + \frac{150}{314.159 \cdot 0.050} (60 - 314.159 \cdot t) \end{aligned}$$

To compute the angle β the following equation must be solved:

$$0 = 300 \left\{ \cos(60) - \cos(\beta) \right\} + 150 (60 - \beta) = f(\beta)$$

A solution can be found starting from $\beta = \pi$ computing some values of $f(\beta)$. After some trials, $\beta = 3.817$ is selected as solution.

Personal notes:

2.3 c

2.3 d

2.3 d

Personal notes:

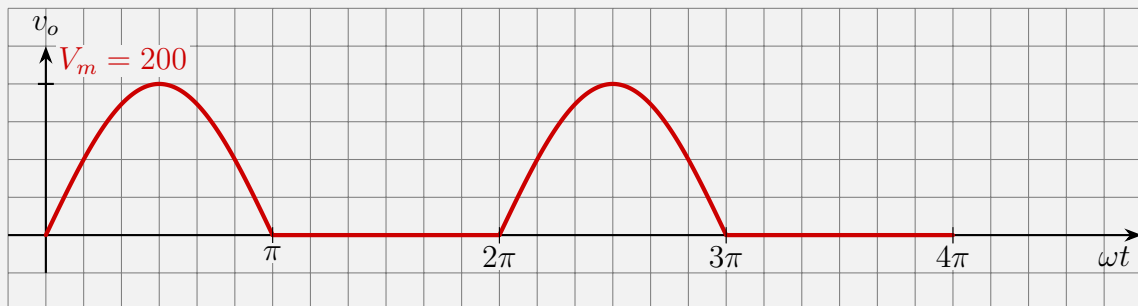
6.4 Result of exercise 2.4 on page 39

2.4 a

The steady state is reached after the $L - R$ load is charged to the average current value. The transient can therefore be estimated as:

$$5\tau = 5 \cdot \frac{L}{R} = 5 \cdot \frac{0.030}{5} = 30.00 \text{ ms}$$

2.4 b



2.4 c

$$V_o = \frac{V_m}{\pi} = \frac{200}{\pi} = 63.66 \text{ V}$$

$$I_o = \frac{V_o}{R} = \frac{200}{5 \cdot \pi} = 12.73 \text{ A}$$

Personal notes:

For the computation of the power it is necessary the rms value of the current which can be computed considering the following table:

n	V_n	Z_n	I_n
0	63.66	5	12.73
1	100.00	10.67	9.37
2	42.44	19.50	2.18
4	8.49	38.03	0.22
6	3.64	56.77	0.06
8	2.02	75.56	0.03
10	1.29	94.38	0.01

The rms value for the current can be approximated as:

$$I_{o,rms} \simeq \sqrt{\sum_{n=0}^{10} I_{n,rms}^2} = 14.44 \text{ A}$$

The power of the load is then:

$$P = R I_{o,rms}^2 = 5 \cdot 14.44^2 = 1042.2 \text{ W}$$

Personal notes:

[illegible]

6.5 Result of exercise 2.5 on page 41

2.5 a

The average voltage and current in the load are:

$$V_o = \frac{V_m}{\pi} = \frac{400}{\pi} = 127.32 \text{ V}$$

$$I_o = \frac{V_o}{R} = \frac{127.32}{4} = 31.83 \text{ A}$$

2.5 b

The ripple current in the load is constrained as:

$$\Delta i_o \leq 0.06 \cdot I_o = 0.06 \cdot 31.83 = 1.91 \text{ A}$$

Such a ripple can be approximated by the first current harmonic:

$$\Delta i_o \approx 2 \frac{V_1}{Z_1} = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}}$$

From which the inductance can be estimated as:

$$L = \frac{V_m}{\Delta i_o \cdot 2\pi f} = \frac{400}{1.91 \cdot 2\pi \cdot 50} = 0.667 \text{ H}$$

2.5 c

Due to the limited value of Δi_o the effect of current ripple on the rms value of the current can be neglected:

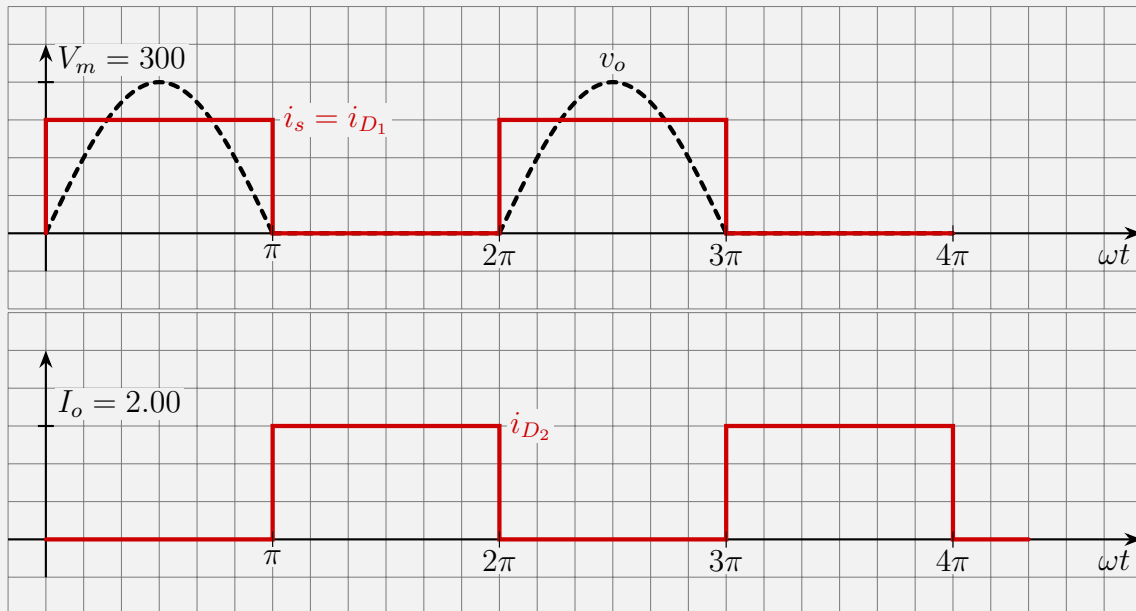
$$I_{o,rms} \simeq I_o = 2.00 \text{ A}$$

The power delivered to the load is then:

$$P = RI_{o.rms}^2 = 2 \cdot 2.00^2 = 3620.25 \text{ kW}$$

Personal notes:

[illegible]



$$V_{s,rms} = \frac{V_m}{\sqrt{2}} = \frac{300}{\sqrt{2}} = 212.13 \text{ V}$$

$$I_{s,rms} = \frac{I_o}{\sqrt{2}} = \frac{2.00}{\sqrt{2}} = 1.41 \text{ A}$$

$$pf = \frac{P}{S} = \frac{P}{V_{s,rms} I_{s,rms}} = \frac{3620.25}{212.13 \cdot 1.41} = 0.97$$

Personal notes:

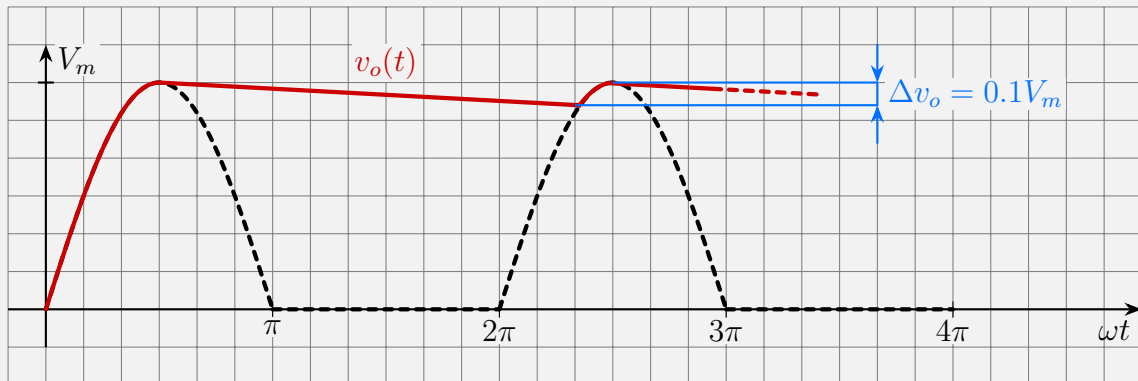
6.6 Result of exercise 2.6 on page 43

2.6 a

The capacitor can be estimated as:

$$C \geq \frac{V_m}{fR\Delta v_o} = \frac{1}{fR\frac{\Delta v_o}{V_m}} = \frac{1}{50 \cdot 100 \cdot 0.1} = 2 \text{ mF}$$

2.6 b



2.6 c

Considering a linear output voltage during the capacitor discharge as a first approximation, the average voltage at the output can be estimated as:

$$V_o = V_m - \frac{\Delta v_o}{2}$$

Considering $\Delta v_o = 0.1V_m$:

$$V_o = V_m - \frac{0.1V_m}{2} = 0.95 \cdot V_m = 308.75 \text{ V}$$

The actual average voltage will be slightly higher than this value.

Personal notes:

6.7 Result of exercise 2.7 on page 45

The output voltage for a controlled rectifier is:

$$V_o = \frac{V_m}{2\pi} (1 + \cos \alpha)$$

The angle α can be computed as:

$$\alpha = \arccos\left(\frac{2\pi}{V_m}V_o - 1\right) = \arccos\left(\frac{2\pi}{300}50 - 1\right) = 1.52 \text{ rad} = 87.3^\circ$$

2.7 a)

The output rms voltage is:

$$V_{o,rms} = \frac{V_m}{2} \sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}}$$

$$= \frac{300}{2} \sqrt{1 - \frac{1.52}{\pi} + \frac{\sin(2 \cdot 1.52)}{2\pi}} = 109.44 \text{ V}$$

and the power:

$$P = \frac{V_{o,rms}^2}{R} = \frac{109.44^2}{100} = 119.7 \text{ W}$$

The rms current in both the load and source is:

$$I_{rms} = \frac{V_{o,rms}}{R} = \frac{109.4}{100} = 1.09 \text{ A}$$

2.7 b)

$$pf = \frac{P}{S} = \frac{119.7}{\frac{300}{\sqrt{2}} \cdot 1.09} = 0.517$$

2.7 c)

Personal notes:

6.8 Result of exercise 2.8 on page 46

2.8 a

For the considered power the required rms voltage is:

$$V_{o,rms} = \frac{V_m}{2} \sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}} = \sqrt{P R} = \sqrt{200 \cdot 100} = 141.42 \text{ V}$$

from which the following expression can be derived for the angle α :

$$\pi \left[\left(\frac{2 V_{o,rms}}{V_m} \right)^2 - 1 \right] = \frac{\sin(2\alpha)}{2} - \alpha = -0.349$$

Solving for α we obtain:

$$\alpha = 0.8454 \text{ rad} = 48.44^\circ$$

α can be computed with some trials ore numerically. For example, the following python snippet could be used:

```
1 from scipy import optimize
2 from math import *
3
4 def f(x):
5     return sin(2*x)/2 - x + 0.349
6
7 alpha = optimize.newton(f, x0=pi)
8 print(f"alpha: {alpha*180./pi}")
```

2.8 b

With a value of $V_m=300$ V the maximum power achievable with a half-wave rectifier is equal to:

$$P_{max} \frac{(V_m/2)^2}{R} = \frac{150^2}{100} = 225 \text{ W}$$

Personal notes:

[illegible]